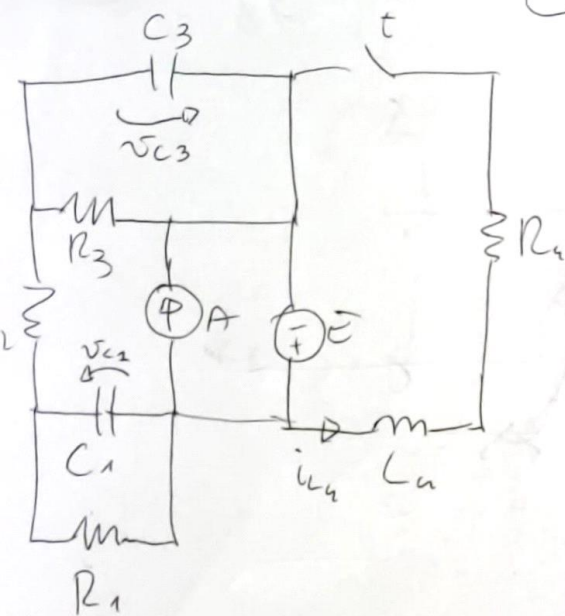


TRANSITORIO



$$R_1 = 2 \Omega, R_2 = 3 \Omega, R_3 = 2 \Omega, R_4 = 3 \Omega,$$

$$A = 1 A, E = 2 V, C_3 = 1 \mu F$$

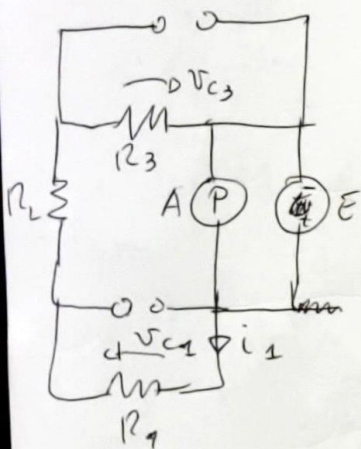
$$L_4 = 4 mH, C_1 = 3 F$$

$$t = 0^- : W_{C1}, W_{C3}, W_{L4}$$

$$t = 0^+ : \frac{dv_{C1}}{dt}, \frac{di_{L4}}{dt}$$

$$t = \infty : Q_{C1}, P_{R4}, P_E$$

$t = 0^-$



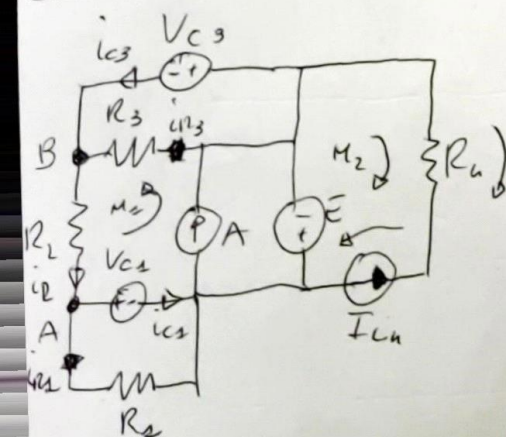
$$i_{L4} = 0 \rightarrow W_{L4} = 0 J$$

$$i_1 = + \frac{E}{R_1 + R_2 + R_3} = + \frac{2}{7} = +0.28 A$$

$$W_{C1} = \frac{1}{2} C_1 v_{C1} = \frac{1}{2} C_1 (-i_1 R_1)^2 = 0.5 mJ$$

$$W_{C3} = \frac{1}{2} C_3 v_{C3} = \frac{1}{2} C_3 (-i_1 R_3)^2 = 0.17 mJ$$

$t = 0^+$



$$LUT(M_1) : -E - v_{C3} + v_{R2} - v_{C1} = 0$$

$$v_{R2} = E + v_{C3} + v_{C1} = -0.84 V$$

$$i_{R2} = \frac{v_{R2}}{R_2} = -0.28 A$$

$$LUC(A) : i_2 - i_{C1} - i_{C3} = 0$$

$$i_{C1} = i_2 - \left(\frac{v_{C1}}{R_1} \right) \approx 0 A$$

$$\frac{dv_{C1}}{dt} = \frac{i_{C1}}{C_1} = 0 \quad \frac{V}{s}$$

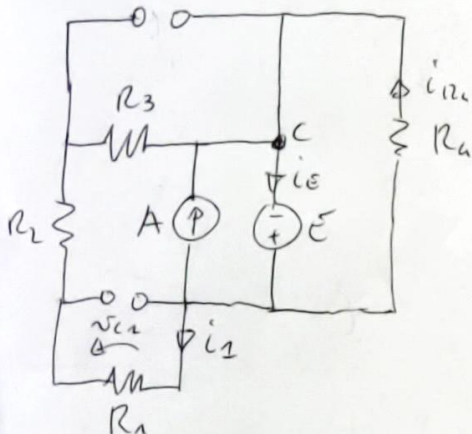
$$v_{R_u} = I_{L_u} \cdot R_u = 0V$$

$$LUT(M_1): v_{R_u} + v_{L_u} - \bar{E} = 0 \rightarrow v_{L_u} = \bar{E} - v_{R_u} = \bar{E} = 2V$$

$$\left[\frac{di_{L_u}}{dt} = \frac{v_{L_u}}{L_u} = 500 \text{ A/s} \right]$$

$$\boxed{t = \infty}$$

$$\boxed{P_{R_u} = \frac{\bar{E}^2}{R_u} = 1.33 \text{ W}}$$



$$i_{R_u} = \frac{\bar{E}}{R_u} = 0.66 \text{ A} \quad i_1 = + \frac{\bar{E}}{R_1 + R_2 + R_3} = 0.20 \text{ A}$$

$$LUC(c): A - i_{\bar{E}} + i_{R_u} + i_1 = 0$$

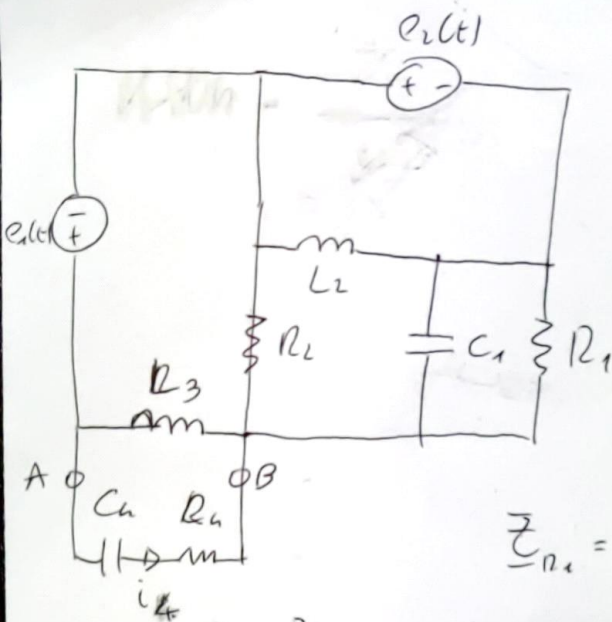
$$i_{\bar{E}} = A + i_{R_u} + i_1 = 1.95 \text{ A}$$

$$\boxed{P_{\bar{E}} = i_{\bar{E}} \bar{E} = 3.9 \text{ W}}$$

$$v_{C_1} = -i_1 R_1 = -0.58 \text{ V}$$

$$\boxed{Q_{C_1} = v_{C_1} C_1 = 1.7 \text{ mC}}$$

THEVENIN



$$e_1(t) = 2\cos(t+30^\circ) \text{ V}$$

$$e_2(t) = 3\cos(t+15^\circ) \text{ V}$$

$$R_1 = 1 \Omega, R_2 = 3 \Omega, R_3 = 2 \Omega,$$

$$C_1 = 2 \text{ mF}, L_2 = 1 \text{ mH}, L_3 = 3 \text{ mH}$$

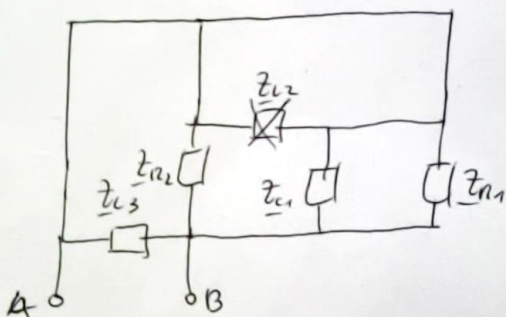
$$C_4 = 4 \text{ mF}$$

$$\underline{Z}_{R_1} = 1 \Omega, \underline{Z}_{R_2} = 3 \Omega, \underline{Z}_{R_3} = 2 \Omega, \underline{Z}_{C_1} = \frac{1}{j\omega C_1} = -500j \Omega,$$

$$\underline{Z}_{L_2} = j\omega L_2 = 0.001j \Omega, \underline{Z}_{L_3} = 0.003j \Omega, \underline{Z}_{C_4} = -250j \Omega$$

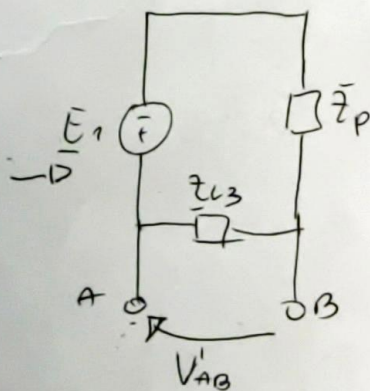
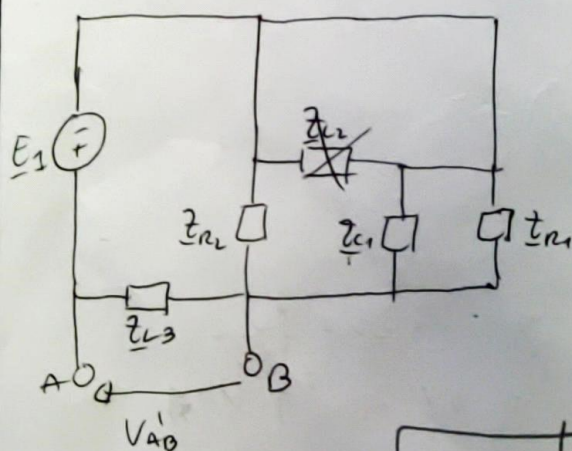
$$\underline{E}_1 = \frac{1}{\sqrt{2}} \angle 30^\circ \text{ V}, \underline{E}_2 = \frac{3}{\sqrt{2}} \angle 15^\circ \text{ V}$$

$\underline{Z}_{eq}$



$$\underline{Z}_{eq} = \underline{Z}_{L_2} \parallel \underline{Z}_{R_1} \parallel \underline{Z}_{C_1} \parallel \underline{Z}_{L_3} \parallel \underline{Z}_{R_2} = 1.2 \cdot 10^{-5} + 3 \cdot 10^{-3}j \Omega$$

$\underline{E}_1 \neq 0$

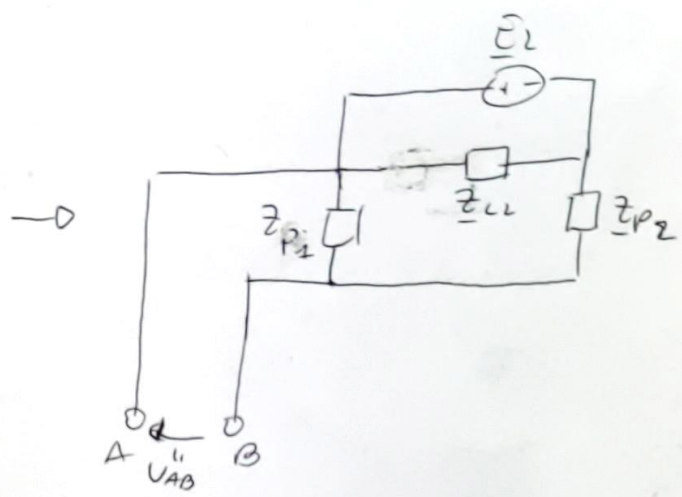
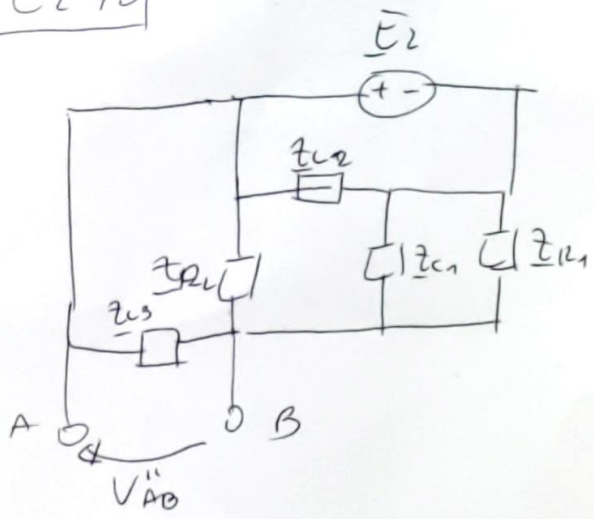


$$\underline{Z}_p = \underline{Z}_{R_1} \parallel \underline{Z}_{C_1} \parallel \underline{Z}_{R_2} =$$

$$= 0.75 - 1 \cdot 10^{-3}j \Omega$$

$$\underline{V}'_{AB} = \underline{E}_1 \frac{\underline{Z}_{L_3}}{\underline{Z}_{L_3} + \underline{Z}_p} = 0.0028 \angle 120^\circ \text{ V}$$

$\bar{E}_2 70$

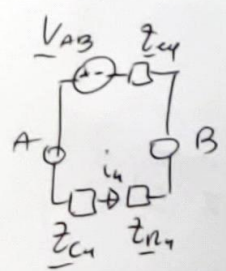


$$z_{P1} = z_{L3} \parallel z_{RL} = 3 \cdot 10^{-6} + 3 \cdot 10^{-3} j \Omega$$

$$z_{P2} = z_{C1} \parallel z_{RL1} = 1 - 2 \cdot 10^{-3} j \Omega$$

$$V_{AB}'' = \bar{E}_2 \frac{z_{P1}}{z_{P1} + z_{P2}} = 63.6 \cdot 10^{-3} \angle 105^\circ \text{ V}$$

$$V_{AB} = V_{AB}' + V_{AB}'' = 9.1 \cdot 10^{-3} \angle 109^\circ \text{ V}$$



$$I_a = \frac{V_{AB}}{z_{L4} + z_{C4} + z_{RL4}} = 3.04 \cdot 10^{-5} \angle 161^\circ \text{ A}$$

$$i_a(t) = 5.15 \cdot 10^{-5} \cos(t - 161^\circ) \text{ A}$$

$$N = z_{L4} |I_a|^2 = (R_{L4} + jX_{L4}) |I_a|^2 = 260.5 - 3312j \text{ VA}$$